

AI-Powered Vehicle Claims Processing

1. Vision

Mission: Help the claims team pay the right amount, as fast as possible, with minimal fraud.

Today, claims agents spend most of their time on information collection, damage review, and cost estimation. These are repeatable tasks that leave little bandwidth for the work that requires human judgment: resolving liability conflicts, managing customer anxiety, and approving complex claims.

This product introduces a multi-agent AI pipeline that absorbs the overhead and surfaces decision-ready claim packages to agents. Every human decision is preserved. AI accelerates the path to it.

Scope note: The core brief targets automation of manual damage review and estimate generation. The AI Intake Agent is an additional assumption that eliminates upstream friction and is scoped as a value-add beyond the core brief

2. User Stories

Claims Agent (Primary)

- As a claims agent, I want to receive a verified, complete claim package so I can focus on assessment, not administration.
- As a claims agent, I want AI-generated damage labels with confidence scores so I can review rather than build assessments from scratch.
- As a claims agent, I want an itemized cost estimate with source attribution so I can approve, adjust, or escalate line items.
- As a claims agent, I want clear fraud risk flags so I can prioritize complex or high-risk claims.
- As a claims agent, I want to give structured feedback on AI outputs so the system improves over time based on my expertise.

Senior Adjuster (Primary)

- As a senior adjuster, I want to review only decision-ready packages so my approval time is minimized.
- As a senior adjuster, I want a full audit trail of AI decisions and human overrides for compliance and accountability.

Policyholder (Secondary)

- As a policyholder, I want to report an incident via a chat interface so I don't have to navigate forms or wait on hold.
- As a policyholder, I want guidance on exactly what photos to submit so my claim isn't delayed.
- As a policyholder, I want immediate confirmation with a reference ID and clear next steps.

3. Key Features

Feature	Description
F1: AI Intake Agent	Conversational mobile interface guiding policyholders through policy verification, incident documentation, and photo submission in a single session. Produces a verified claim object handed to the assessment pipeline. Acts as a quality gate for F2 and F3. Alternatives to be discussed during interview.
F2: Damage Labeller Agent	Computer vision model that analyzes damage photos and generates bounding boxes, part identification, severity classification, and confidence scores per label. Auto-escalates below configurable confidence threshold.
F3: Cost Estimator Agent	Fine-tuned LLM with RAG over repair cost database. Takes approved damage labels and generates an itemized estimate with source attribution per line item. Flags total loss vs. repair gate for human decision.
F4: Claims Agent Dashboard	State-driven web interface surfacing decision-ready claim packages. Next required action always prominently displayed. All AI outputs are transparent, editable, and overridable at line-item level.
F5: Fraud Detection Agent	Parallel, non-blocking multimodal classifier scoring claims Low/Medium/High using submitted photos, incident metadata, and claims history. High risk holds claim for agent review before assessment begins.
F6: Senior Adjuster Review Interface	Condensed approval interface surfacing agent-approved estimates for final sign-off. Structured rejection reasons routed automatically back to claims agent with notification.
F7: Continuous Improvement Layer	Feedback capture layer intercepting all human approvals, adjustments, and overrides. Structures them as labelled retraining signals automatically. No manual retraining pipelines required.
F8: Policyholder Comms	AI-generated claim confirmation, estimate summary, repair shop mapping, and next steps. Agent reviews before sending.

F1 Dependency note: The Intake Agent is not scored independently via RICE. Its value is measured by how much it improves input quality for F2 and F3. Poor photo quality or incomplete incident data degrades labeller and estimator performance across all claims. F1 is infrastructure for the assessment pipeline, not a standalone feature.

4. Prioritization

Features scored using RICE framework against the 3 mission objectives: faster TAT, right amount, and minimal fraud.

Feature	Reach	Impact (1-3)	Confidence	Effort (1-5)	RICE Score	Phase
F2: Damage Labeller Agent	100% Claims	3	80%	3	80	MVP
F3: Cost Estimator Agent	100% Claims	3	80%	4	60	MVP
F4: Claims Dashboard	100% Claims	2	90%	2	90	MVP
F6: Adjuster Review	100% Claims	1	90%	1	90	MVP
F5: Fraud Detection	10-20% of claims	3	60%	5	7	V1
F8: Policyholder Comms	100% Claims	<1	80%	1	~50	V1
F7: Continuous Improvement	100% Claims	3	70%	5	42	V2

F1 (Intake Agent) excluded from RICE scoring. See dependency note in Section 3. Included in MVP as quality infrastructure for F2 and F3.

5. Success Metrics

Industry baselines used as proxies. To be replaced with client actuals post-pilot. These are business metrics (lagging)

Metric	Baseline	Target
Time from FNOL to estimate generated	5-7 days (J.D. Power, 2024)	<24 hours
Damage assessment accuracy vs. settled amount	Unknown	Within 10% variance
Claims agent time on intake and assessment	~60% of work time (proxy)	<20% by V2
Fraud flag precision (true positive rate)	Manual, inconsistent	>75% by V1
Policyholder CSAT post-interaction	871/1000 digital avg (J.D. Power)	Maintain or improve
Claim intake completion rate (first submission)	~60% (industry proxy)	>90%

AI-Specific Metrics (leading metrics)

- Labeller agent: precision/recall on damage ID, confidence score distribution, human override rate.
- Cost estimator: line item acceptance rate, estimate variance vs. repair shop final price, RAG retrieval precision.
- Fraud agent: true positive rate, false positive rate (under 10% threshold to avoid friction on legitimate claims).
- Intake agent: claim completion rate on first submission, CSAT, LLM-as-judge empathy score.

6. AI Integration

Architecture Overview

Four specialized agents operate as a sequential pipeline with a parallel fraud detection layer. Each agent has discrete inputs, outputs, confidence scoring, and an independent feedback loop.

Agent	Model Type	Inputs/Tools	Outputs	Human Trigger
Damage Labeller Agent	Computer vision (fine-tuned)	Damage photos	Bounding boxes, part ID, severity, confidence score per label	Below confidence threshold, rare vehicle model, overlapping damage
Cost Estimator Agent	Fine-tuned LLM + RAG	Approved labels, vehicle data, repair cost DB	Itemized estimate with source attribution, total loss flag	All estimates: claims agent approves at line-item level
Intake Agent	Conversational LLM	Policyholder messages, policy DB	Verified incident object, claim request object	Invalid policy, injury, liability conflict
Fraud Detection Agent	Multimodal classifier + anomaly detection	Photos, incident metadata, claims history	Fraud risk score (Low/Med/High) + contributing signals	High risk: claim held for agent review

Human-AI Interaction Model

AI is never a gatekeeper. Every claim reaches a human reviewer. The interaction model operates on three levels:

- Approve: Claims agent accepts AI output. Logged as positive training signal.
- Adjust: Claims agent modifies at line-item level. Delta logged as correction signal.
- Override: Claims agent rejects entirely. Reason logged as strong retraining signal.

Confidence thresholds are configurable per agent and tighten over time as feedback accumulates.

A dedicated feedback capture layer (Scale AI Dialect) intercepts all human decisions, extracts reasoning, and structures them as labelled inputs for automatic model improvement.

Observability Principle

Every AI inference is traceable to its source. Cost estimate line items link to the repair DB entry or historical claim that informed them. Damage labels link to confidence signal and training data category. Fraud flags link to contributing signals. No AI output is presented without a visible reasoning trail.

Edge Case Handling

- Overlapping or complex damage: labeller escalates to claims agent with partial output as starting point.
- Rare vehicle models: escalated immediately with low-confidence flag.
- Total loss determination: flagged by cost estimator, mandatory human decision.
- High fraud risk: claim held before assessment pipeline begins.

7. Legal Considerations and Ethical Risks

Legal Liability

- The insurer owns every AI-assisted decision. Liability never transfers to the AI vendor or the model.
- Mandatory human approval at every decision gate is a legal requirement, not just a UX choice.
- Audit trail of all AI outputs and human decisions must be maintained for dispute resolution and regulatory compliance.

Ethical Risks

Model bias from skewed training signals: If human feedback is driven by cost minimization rather than accuracy, the model learns to minimize payouts, not to be correct.

Discriminatory fraud flagging: If historical fraud data carries demographic bias, the fraud model may systematically disadvantage certain policyholders. Regular bias audits across demographic segments are required before and after V1

Privacy: Submitted photos contain PII including faces, license plates, and location metadata. Claims history is sensitive personal data. Redaction pipeline is mandatory before any data enters the training pipeline. Data usage must comply with applicable regulations per jurisdiction.

Opacity: Policyholders have a right to understand why a claim was flagged, adjusted, or rejected. A structured explanation and appeal path must exist for every AI-influenced decision.
